DES Pune University, Pune
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Title:	**To write a C++ program to add a virtual function**	
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Contents

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Theory (Logic)

This program shows how polymorphism works in C++ using virtual functions. Classes: Base class (base): Has a function print() that prints "print base class". It is marked as virtual, which means if a derived class has the same function, it can override it at runtime. Derived class (derived): Inherits from base. Has its own version of print() which prints "print derived class". In main():A pointer b of type base* is declared. An object d of type derived is created. The pointer b is assigned the address of d (b = &d). Then: d.print(); calls the derived class version directly. b->print(); calls the derived version because print() is a virtual function. If print() was not marked virtual in the base class, it would call the base version instead.

Pseudo Code

- **Step 1: Include input-output library**
- **Step 2: Use standard namespace**
- **Step 3: Create class base**
    - **Public:**
    - **Virtual function print()**
    - **Print "print base class"**
- **Step 4: Create derived class**
    - **Public:**
    - **Function print()**
    - **Print "print derived class"**
- **Step 5: Start main function**
    - **Create pointer b of type base***
    - **Create object d of type derived**
    - **Set b to point to d**
    - **Call print() using d**
    - **Call print() using pointer b**
- **Step 6: End main function**

Program Code

```
#include<iostream>
using namespace std;
class base
{
   public:
   virtual void print()
   {
      cout<<"print base class\n";
   }
};

class derived:public base
{
   public:
   void print()
   {
      cout<<"print derived class\n";
   }
};

int main()
{
   base* b;
   derived d;
   b= &d;
   d.print();
   b->print();
   return 0;
}
```

Output

```
print derived class
print derived class
```

Conclusion

Code is run successfully, required output is received.